

CS2009 - Design and Analysis of Algorithm

Fall 2024

Project Description

Due Date: 20th Nov 2024

Total Marks: 100 (Weightage 10)

You need to work as a Group of 3

In this project, you are required to implement the following problems using closest points algorithms with various Big Oh complexities. User can either draw objects in screen or you can also design best way to take input from user e.g. through files. You need to show a very nice user interface to display steps in algorithms and all various steps of algorithms must be clearly shown. In addition, you need to show steps how to calculate Time and Space complexities of each algorithm.

- Given n points in a two dimensional space, find the closest points. Use brute force algorithm to find the closest points [25 Points]
- Given n points in a two dimensional space find the closest point. use divide and conquer algorithm discussed in class, to find the closest points [25 Points]
- You have to research for divide and conquer closest point algorithm for more than two dimensional space (From research papers). Demonstrate your implementation for 3D points and clearly cite the paper [20 Points through Demo]

You are required to submit a 2-page report (pdf must be generated through LaTeX) with the following sections with 1-page maximum as Appendix [Times New Roman, 2 Columns and 10 font size] [30 Points]

- (a) Abstract
- (b) Introduction
- (c) Your programming design (Programming Language used etc, make a clear diagram of your system here along with discussion)
- (d) Experimental Setup
- (e) Results and Discussion (Here, show screen shots etc along with discussion) You can also show execution time etc and compare various algorithms)
- (f) Conclusion
- (g) References